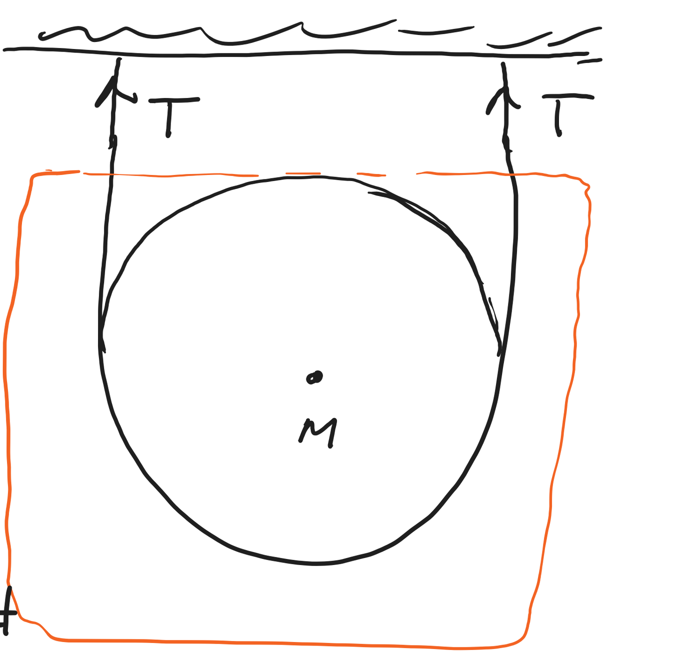
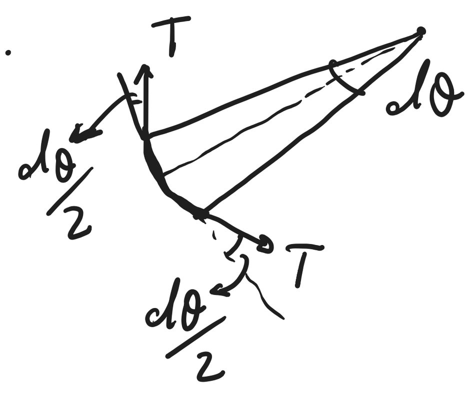
Problem 1)

a) Isolate the pulley:



$$2T = mg \Rightarrow \boxed{T = \frac{mg}{2}}$$

To obtain the normal force, consider a tiny element of the pulley subtending $d\theta$.



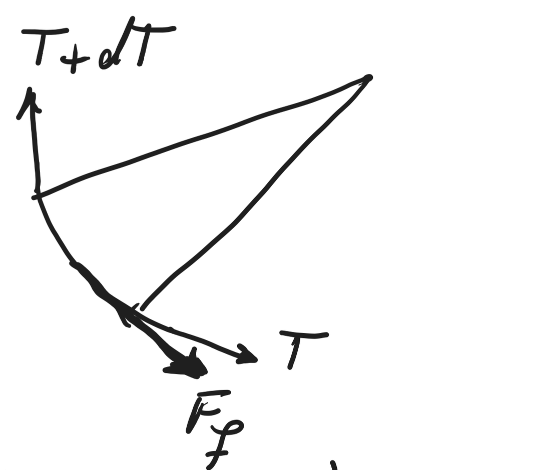
The tension is identical at both ends because string is massless and frictionless.

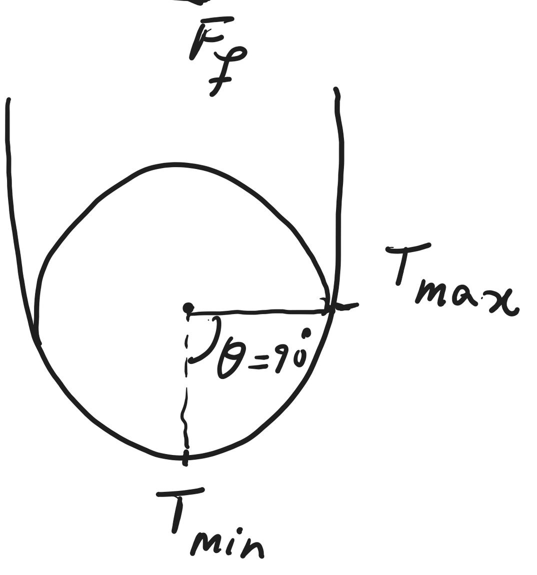
The normal force N is in response to the inward components of these two tangential tensions:

$$T_{inward} = T \sin\frac{d\theta}{2} \cong T\frac{d\theta}{2} \Rightarrow dN = N d\theta = 2T\frac{d\theta}{2}$$

$$\Rightarrow \boxed{N = \frac{mg}{2}} \Rightarrow \boxed{\frac{N}{R} = \frac{mg}{2R}}$$

(b) Because of friction, the tension is now differential. Thus, the lowest value of tension occurs at the lowest point of the pulley.





$$\left.\begin{aligned} T + F_f &= T + dT \\ F_f &= \mu\, dN = \mu T d\theta \end{aligned}\right\} \Rightarrow dT = \mu T d\theta$$

$$\Rightarrow \int_{T_{min}}^{T_{max}} \frac{dT}{T} = \mu \int_0^{\pi/2} d\theta \quad ; \quad T_{max} = \frac{mg}{2}$$

$$\Rightarrow \ln T \Big|_{T_{min}}^{T_{max}} = \mu\frac{\pi}{2} \Rightarrow \frac{T_{max}}{T_{min}} = e^{\mu\frac{\pi}{2}}$$

$$\Rightarrow \boxed{T_{min} = \frac{mg}{2} e^{-\mu\frac{\pi}{2}}}$$

Remark: $\mu \to 0 \Rightarrow \boxed{T_{min} \geqslant \frac{mg}{2}}$ which makes sense from part (a).

$\mu \to \infty \Rightarrow \boxed{T_{min} \geqslant 0}$

Problem 2)

known: v_0 init. velocity
$\vec{F} = -km\vec{v}$

unknown: prove $v = v_0 - kx$

$$F = ma = m\frac{dv}{dt} = mv\frac{dv}{dx} = -km\vec{v}$$

$$\Rightarrow dv = -k\,dx \Rightarrow \int_{v_0}^{v(t)} dv = -k\int_0^{x(t)} dx$$

$$\Rightarrow \boxed{v(t) = v_0 - k\,x(t)}$$

Problem 3)

$$r = x\hat{x} + y\hat{y} + z\hat{z}$$
$$v = \dot{x}\hat{x} + \dot{y}\hat{y} + \dot{z}\hat{z}$$
$$a = \ddot{x}\hat{x} + \ddot{y}\hat{y} + \ddot{z}\hat{z}$$

$\vec{B} = B_0\hat{y}$

$\vec{F} = q\vec{v}\times\vec{B} = m\vec{a}$

(x_0, y_0, z_0) is starting point.

$$\Rightarrow m\left(\ddot{x}\hat{x} + \ddot{y}\hat{y} + \ddot{z}\hat{z}\right) = q\left(\dot{x}\hat{x} + \dot{y}\hat{y} + \dot{z}\hat{z}\right)\times\left(B_0\hat{y}\right)$$
$$= qB_0\left(\dot{x}\hat{z} - \dot{z}\hat{x}\right)$$

$$\Rightarrow \begin{cases} m\ddot{x}\hat{x} = -qB_0\dot{z}\hat{x} \\ m\ddot{y}\hat{y} = 0 \longrightarrow \ddot{y} = 0 \Rightarrow \dot{y} = \dot{y}_0 = \text{const.} \Rightarrow \boxed{y = \dot{y}_0 t + y_0} \\ m\ddot{z}\hat{z} = +qB_0\dot{x}\hat{z} \end{cases}$$

For simplicity: $\gamma = \frac{qB_0}{m}$

$$\left.\begin{cases} \ddot{x} = -\gamma\dot{z} \Rightarrow \dddot{x} = -\gamma\ddot{z} \\ \ddot{z} = \gamma\dot{x} \Rightarrow \dddot{z} = \gamma\ddot{x} \end{cases}\right\} \rightarrow \text{Substitute the later eqns. in the former eqns.}$$

$$\Rightarrow \begin{cases} \frac{\dddot{z}}{\gamma} = -\gamma\dot{z} \\ -\frac{\dddot{x}}{\gamma} = \gamma\dot{x} \end{cases} \Rightarrow \begin{cases} \dddot{z} = -\gamma^2\dot{z}, \text{ let } \omega = \dot{z} \\ \dddot{x} = -\gamma^2\dot{x}, \text{ let } \eta = \dot{x} \end{cases} \Rightarrow \begin{cases} \ddot{\omega} = -\gamma^2\omega \\ \ddot{\eta} = -\gamma^2\eta \end{cases}$$

$$\Rightarrow \begin{cases} \omega = \alpha_\omega \sin(\gamma t) + \beta_\omega \cos(\gamma t) \\ \eta = \alpha_\omega \sin(\gamma t) + \beta_\omega \cos(\gamma t) \end{cases} \Rightarrow \begin{cases} z = \alpha_z \sin(\gamma t) + \beta_z \cos(\gamma t) + z_0 \\ x = \alpha_x \sin(\gamma t) + \beta_x \cos(\gamma t) + x_0 \end{cases}$$

But recall $\alpha_x, \alpha_z, \beta_x, \beta_z$ are connected through eqns above:

$$\begin{cases} \ddot{x} = -\alpha\dot{z} \\ \ddot{z} = \alpha\dot{x} \end{cases} \Rightarrow \begin{cases} \alpha_z = \beta_x = \alpha \\ \alpha_x = -\beta_z = \beta \end{cases} \quad \begin{cases} x - x_0 = \alpha\cos(\gamma t) + \beta\sin(\gamma t) \\ y - y_0 = \dot{y}_0 t \\ z - z_0 = -\beta\cos(\gamma t) + \alpha\sin(\gamma t) \end{cases}$$

The initial conditions, $t = 0$, $\dot{z}(t=0) = \dot{z}_0$; $\dot{x} = 0$ further yield,

$$\begin{cases} \gamma\beta = 0 \\ \gamma\alpha = \dot{z}_0 \end{cases} \Rightarrow \boxed{\begin{cases} x - x_0 = \frac{\dot{z}_0}{\gamma}\cos(\gamma t) \\ y - y_0 = \dot{y}_0\, t \\ z - z_0 = \frac{\dot{z}_0}{\gamma}\sin(\gamma t) \end{cases}}$$